2 The Coriolis Force

The acceleration in a rotating frame of reference differs from the acceleration in an inertial frame by two terms, as derived below for a particle moving with velocity relative to the rotating Earth.

$$\frac{d\mathbf{u}}{dt} + 2\,\boldsymbol{\Omega} \times \mathbf{u} = -\frac{1}{\rho}\nabla p + \mathbf{g} \tag{2.1}$$

The second term on the left is the Coriolis acceleration, and the centrifugal term has been absorbed into the effective gravity. Written in components on the tangent plane, the momentum equations become

$$\frac{\partial u}{\partial t} - f v = -\frac{1}{\rho_0}\frac{\partial p}{\partial x} \tag{2.2}$$

$$\frac{\partial v}{\partial t} + f u = -\frac{1}{\rho_0}\frac{\partial p}{\partial y} \tag{2.3}$$

where the Coriolis parameter is defined from the rotation rate and the latitude of the tangent plane as

$$f = 2\,\Omega \sin\varphi, \qquad \tilde{u} = \bar{u} + u' \tag{2.4}$$

For typical midlatitude values the inertial period is about seventeen hours, which sets the time scale over which rotation matters for oceanic and atmospheric motions.